

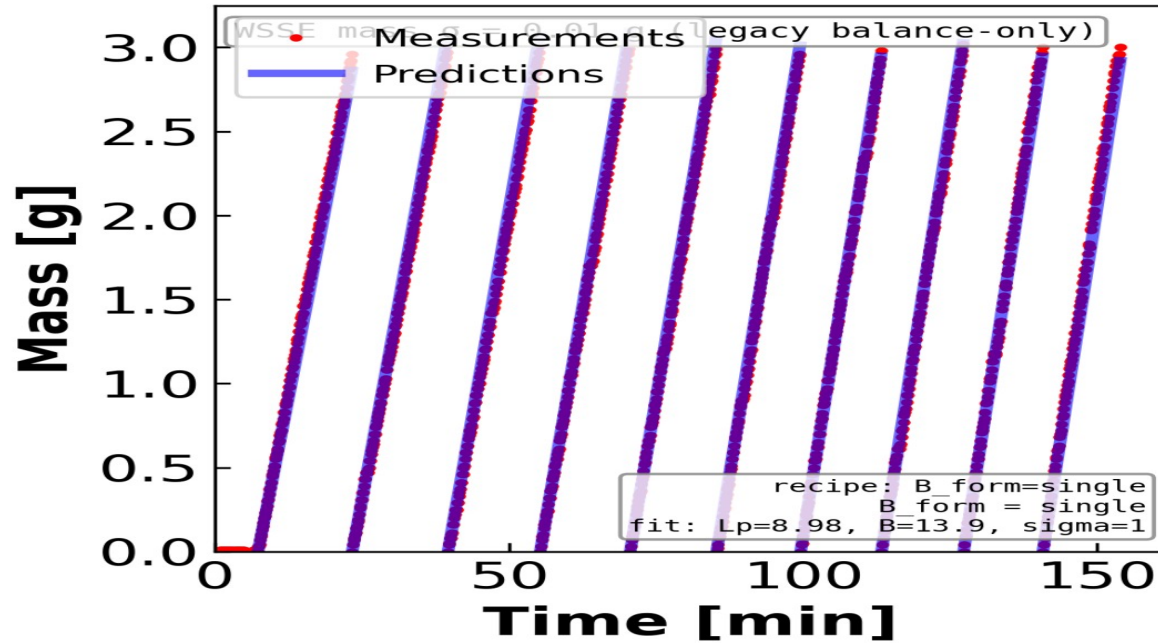
Fitting the four representative DATA3 NF270 experiments

Measured vs predicted at the optimized parameters — NaCl (diluting & concentrating), CaCl₂, LaCl₃

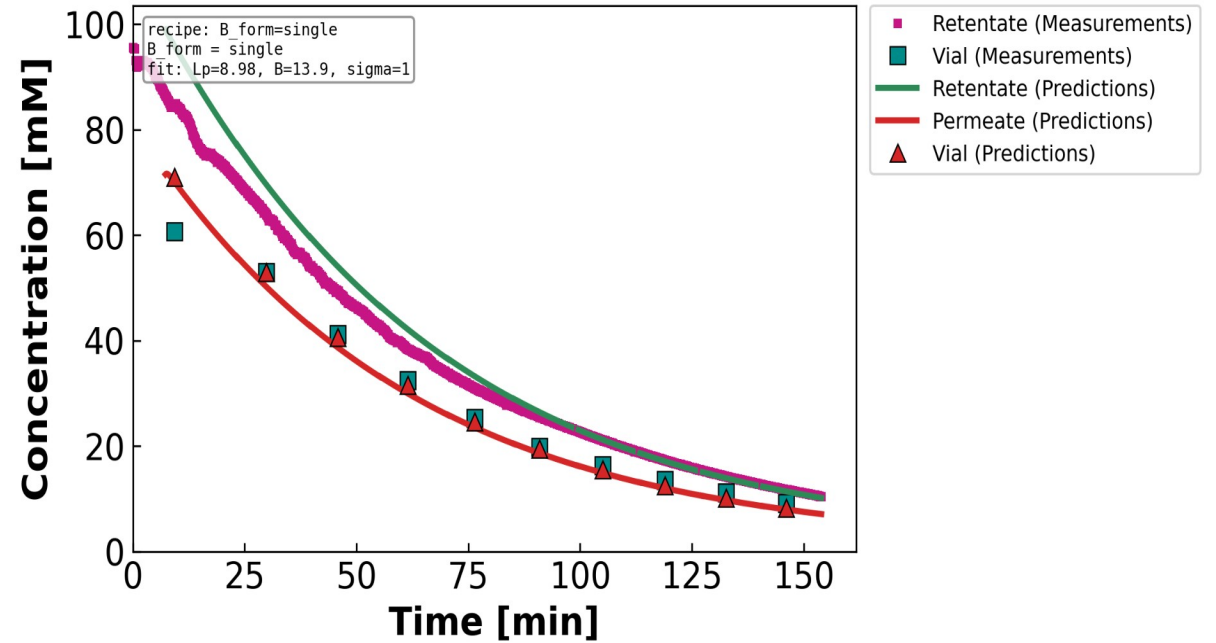
NaCl — diluting regime (MC3 · 07.22.24)

fit form: constant B

Mass — measured (points) vs predicted (line)



Concentration — retentate / permeate



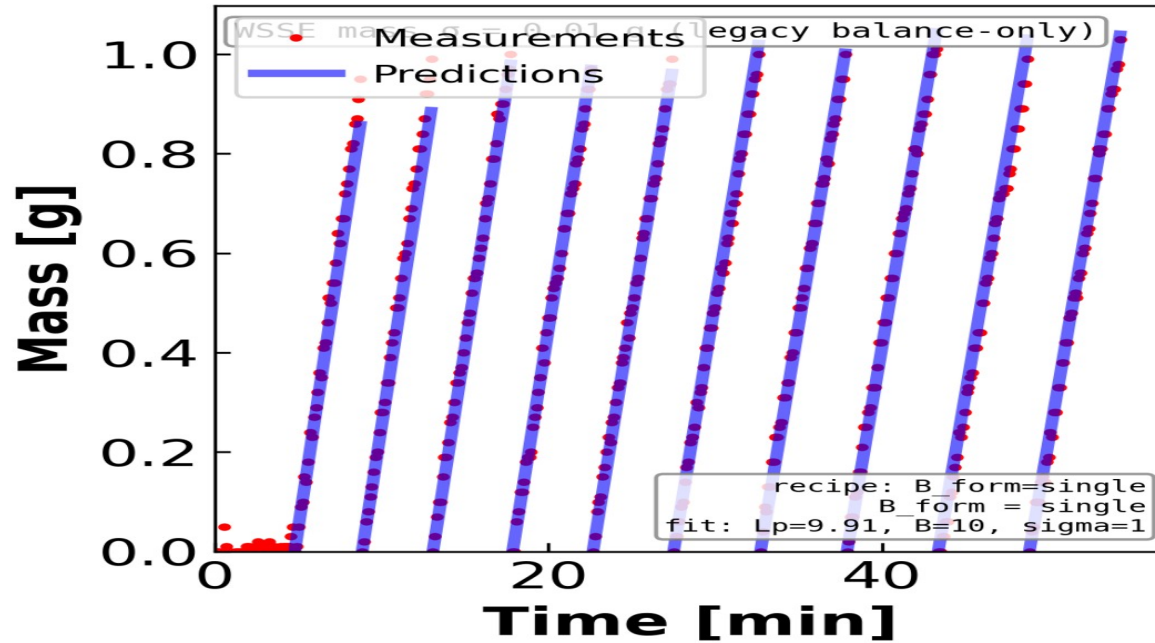
Optimized parameters
 $L_p = 8.98 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$
 $B = 13.88 \text{ } \mu\text{m s}^{-1}$
 $\sigma = 1.00$
 $\text{WSSE}_3 = 42$

→ B is flat over the range — constant B already tracks mass and both concentrations ($\text{WSSE}_3=42$).

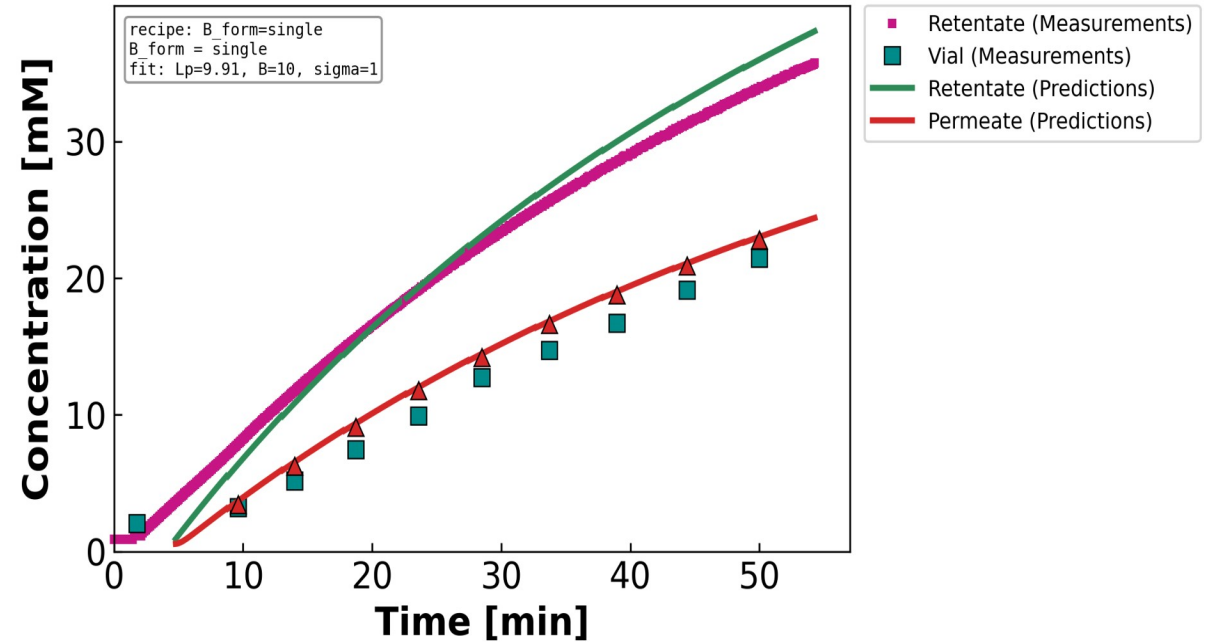
NaCl — concentrating regime (MC2 · 05.07.24)

fit form: constant B

Mass — measured (points) vs predicted (line)



Concentration — retentate / permeate



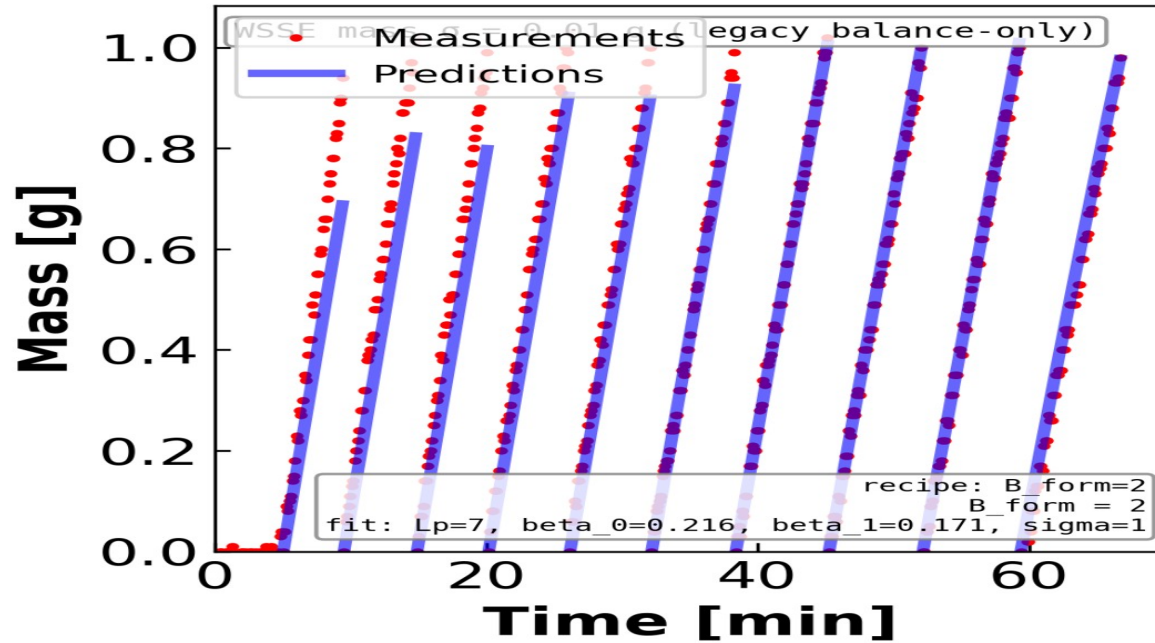
Optimized parameters
 $L_p = 9.91 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$
 $B = 10.00 \text{ } \mu\text{m s}^{-1}$
 $\sigma = 1.00$
 $\text{WSSE}_3 = 114$

→ Constant B is adequate for NaCl ($\text{WSSE}_3=114$); B is essentially concentration-independent.

CaCl₂ (MC2 · 05.07.24)

fit form: quadratic $B = Jw(\beta_0 + \beta_1 c + \beta_2 c^2)$

Mass — measured (points) vs predicted (line)



Optimized parameters

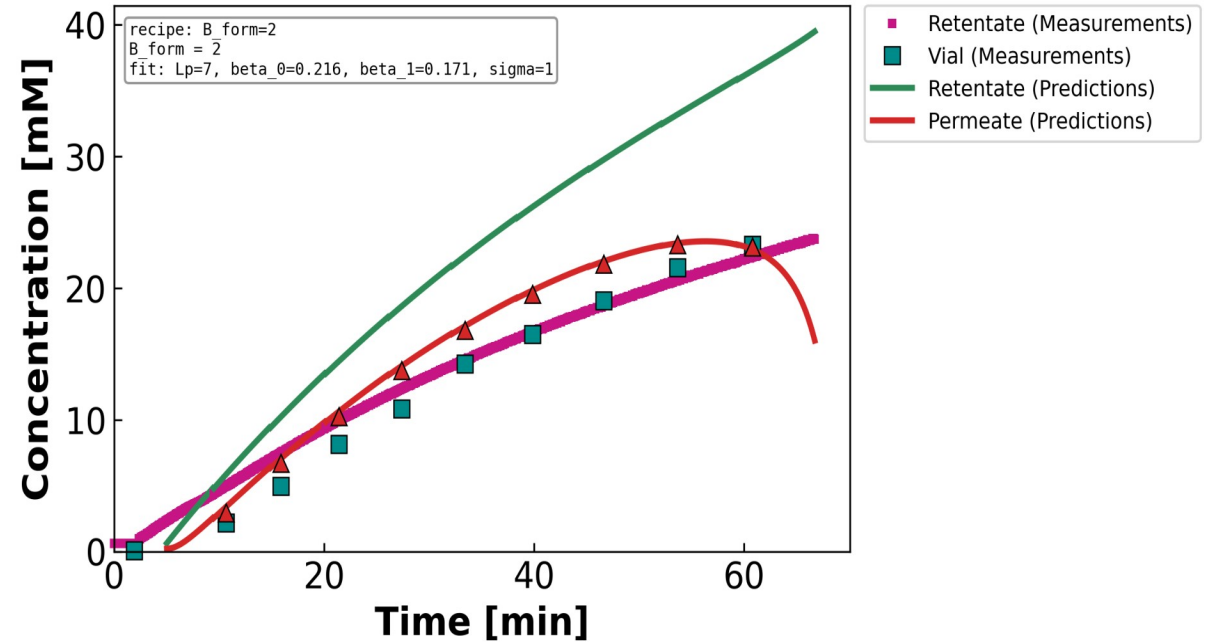
$L_p = 7.00 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

$\beta_0 = 0.216$ $\beta_1 = 0.171$ $\beta_2 = -0.0035$

$\sigma = 1.00$

WSSE₃ = 147

Concentration — retentate / permeate

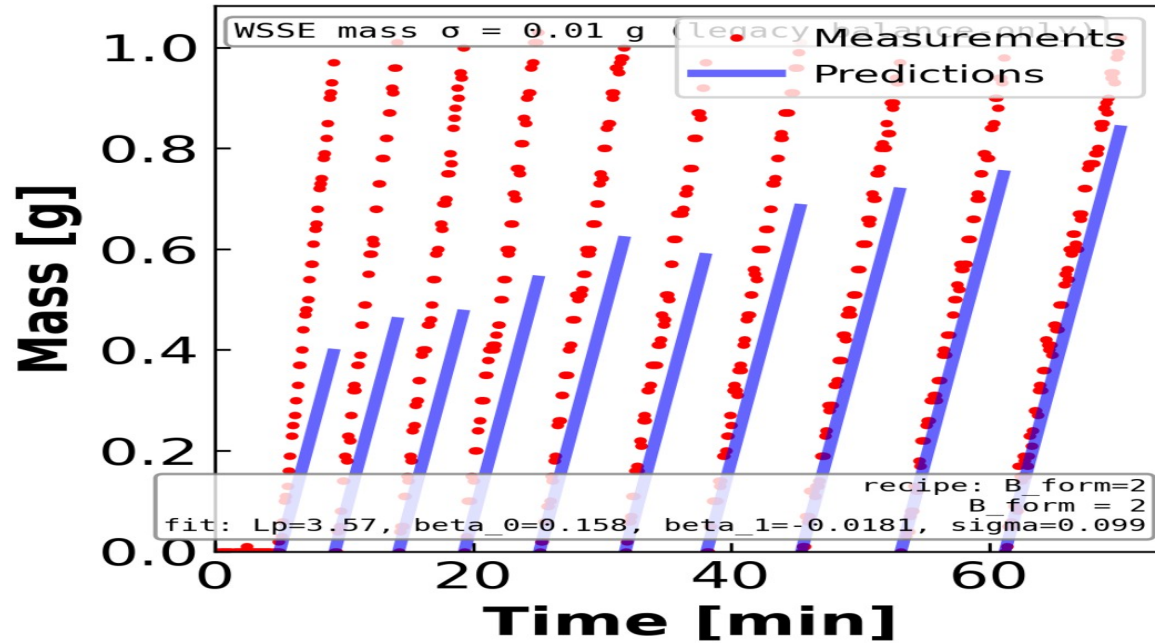


→ B(c) is required: quadratic tracks the permeate (direct solute signal) and cuts WSSE₃ 220→147 (~33%); the retentate is still over-predicted at high c → $\sigma(c)$ /polarization beyond B(c).

LaCl₃ (MC2 · 05.21.24)

fit form: quadratic $B = Jw(\beta_0 + \beta_1 c + \beta_2 c^2)$

Mass — measured (points) vs predicted (line)



Optimized parameters

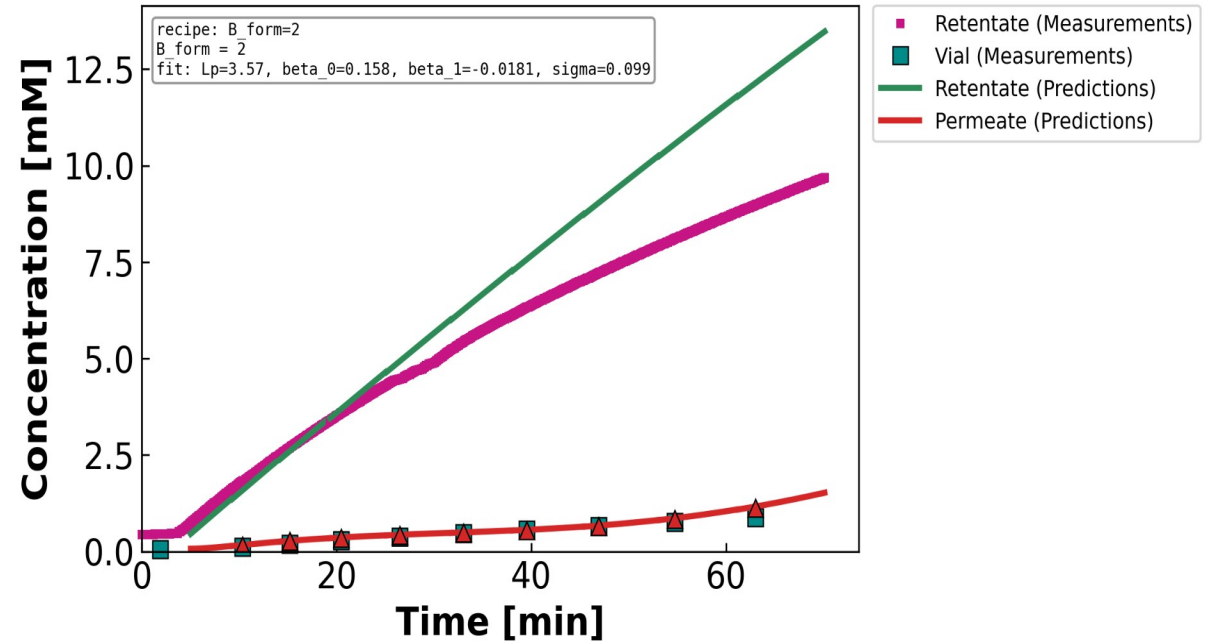
$L_p = 3.57 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$

$\beta_0=0.158$ $\beta_1=-0.018$ $\beta_2=0.0009$

$\sigma = 0.10$

$WSSE_3 = 1192$

Concentration — retentate / permeate

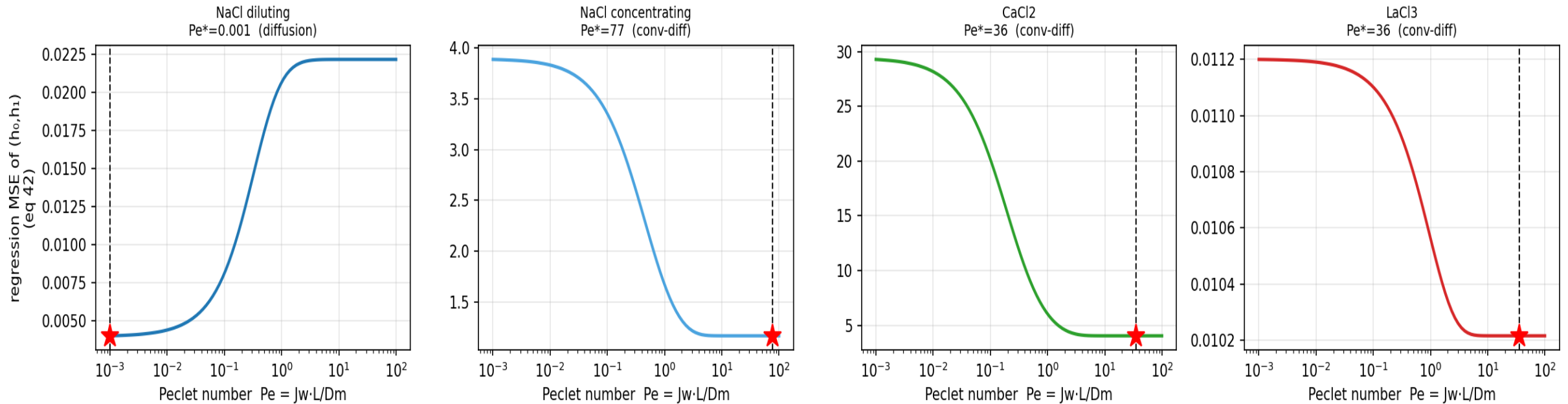


→ Model-limited: no B(c) form captures the trivalent — mass under- and retentate over-predicted ($WSSE_3 \sim 1190-1244$); the residual points past B(c) alone ($\sigma(c)$, ion-pairing).

Why B depends on concentration — the Peclet regime

Regression MSE vs Pe ($\triangleq$ DATA2 Fig 8): which transport regime each salt sits in

DATA3 Peclet analysis ($\triangleq$ DATA2 Fig 8A): regression MSE vs Pe — a minimum at $Pe \geq 1$ means convection+diffusion, i.e. B is concentration-dependent (not the $Pe \rightarrow 0$ constant-B limit)

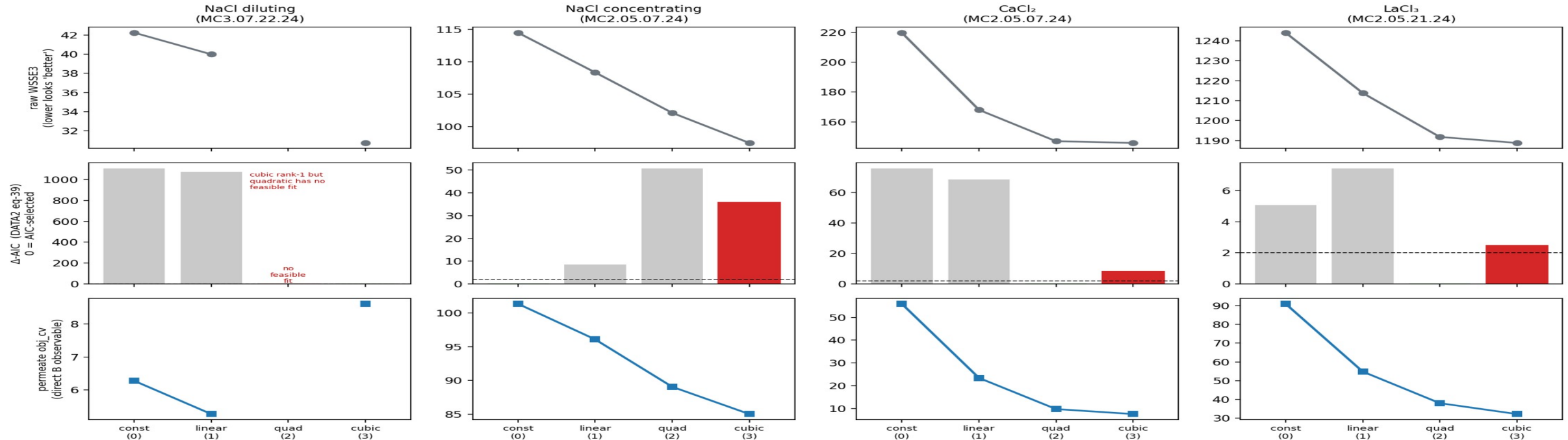


→ Diluting NaCl → $Pe \rightarrow 0$ (diffusion-limited, $B = D_m H / L = \text{constant}$). CaCl₂ / concentrating-NaCl / LaCl₃ → $Pe \gg 1$ (convection+diffusion → $B = Jw \cdot \sum \beta_i c^i$). The Peclet number is WHICH salts need a concentration-dependent B.

Which B(c) order — the DATA2 per-response AIC (not raw fit)

Order chosen by the per-channel AIC (eq 39); raw WSSE3 always over-selects more parameters

DATA3 B(c) order selection — DATA2 per-response AIC, not raw WSSE3. Top: raw WSSE3 always falls with order (the overfit trap). Middle: Δ -AIC selects low order (green) and rejects cubic (red) wherever a full candidate set converged; hatched = no feasible fit (cubic 'wins' on the two diluting NaCl sheets only' because quadratic is missing). Bottom: the permeate channel (the direct B observable) collapses for CaCl_2 = real low-order B(c); it barely moves for NaCl = B(c) weak.

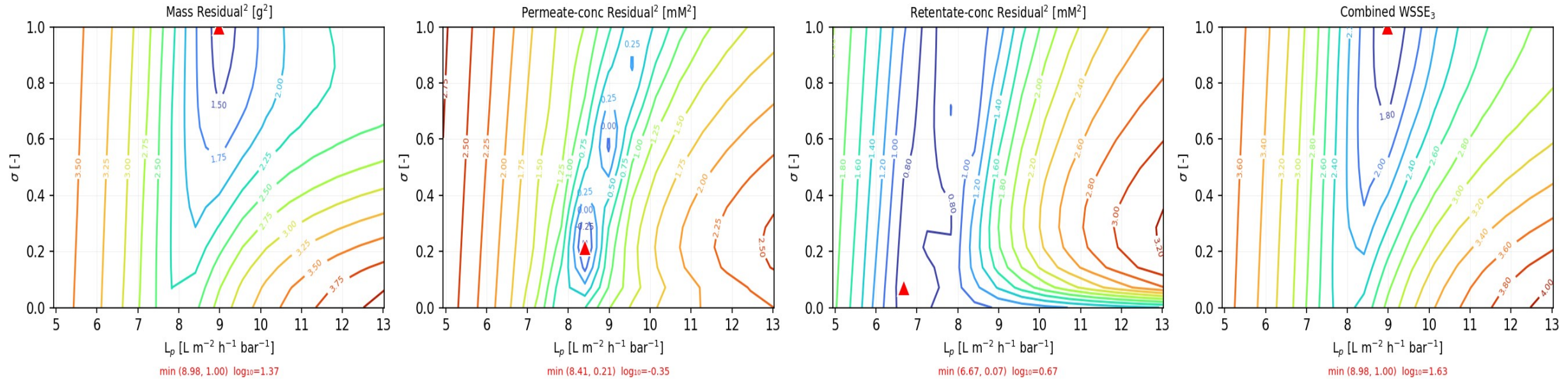


→ CaCl_2 → quadratic / linear, cubic REJECTED ($\Delta=8-25$); NaCl B is flat (constant). The per-response AIC is the honest criterion — the pooled single-n AIC stored in the JSONs over-picks cubic.

Supporting contour — NaCl — diluting regime

$\sigma \times L_p$ objective contour $\cdot$ constant B

MC3.07.22.24_SNaCl $\cdot$ B-form: single $\cdot$ profile contour ($L_p \times \sigma$, B(c) re-solved at each node)



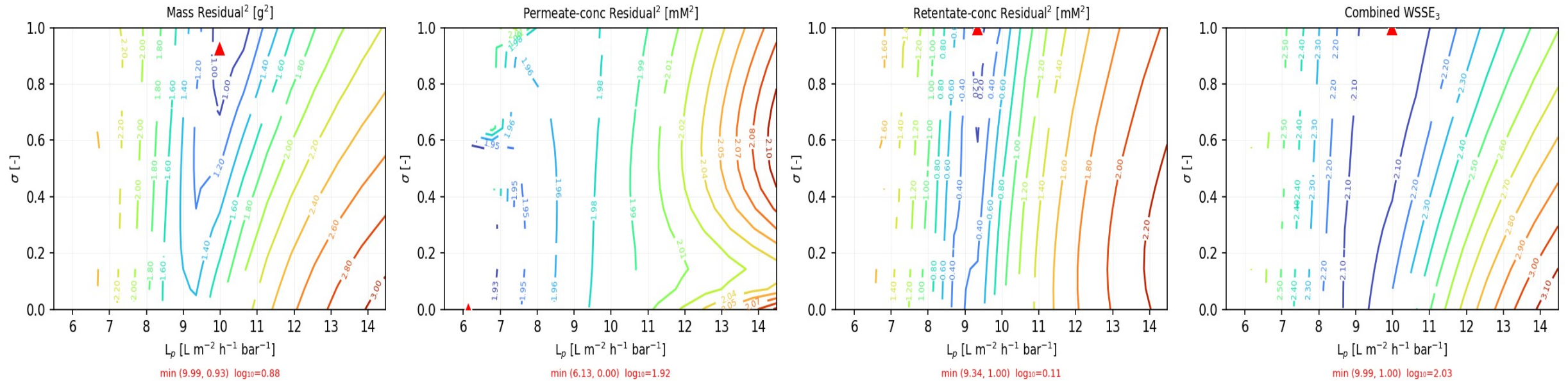
Objective landscape per channel (mass | permeate | retentate | combined), other params re-solved at each node; red $\blacktriangle$ = minimum. A sharp minimum = identified; a flat valley along σ = the σ -rail.

→ clear L_p minimum; σ rails (flat valley) but B is flat so the rail is harmless.

Supporting contour — NaCl — concentrating regime

$\sigma \times L_p$ objective contour · linear B(c) (nearest available form/plane)

MC2.05.07.24_NaCl · B-form: poly1 · profile contour ($L_p \times \sigma$, B(c) re-solved at each node)



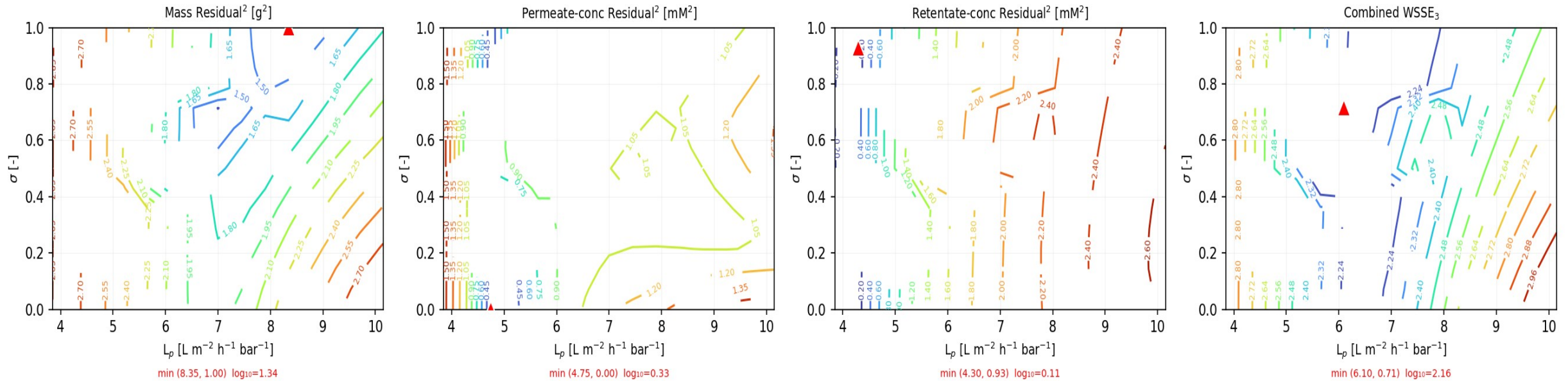
Objective landscape per channel (mass | permeate | retentate | combined), other params re-solved at each node; red ▲ = minimum. A sharp minimum = identified; a flat valley along σ = the σ -rail.

→ constant-B identifiable in B- L_p ; σ rails — adequate because B is concentration-independent.

Supporting contour — CaCl_2

$\sigma \times L_p$ objective contour • quadratic $B(c)$

MC2.05.07.24_ CaCl_2 • B-form: poly2 • profile contour ($L_p \times \sigma$, $B(c)$ re-solved at each node)



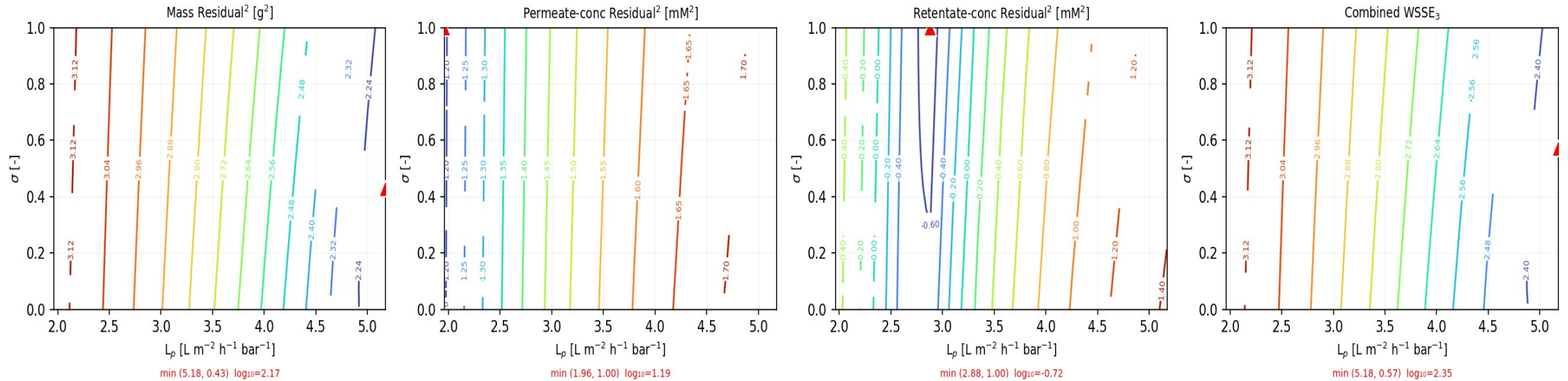
Objective landscape per channel (mass | permeate | retentate | combined), other params re-solved at each node; red $\blacktriangle$ = minimum. A sharp minimum = identified; a flat valley along σ = the σ -rail.

→ with quadratic $B(c)$ profiled, the permeate/retentate channels pin the landscape — the $B(c)$ win.

Supporting contour — LaCl_3

$\sigma \times L_p$ objective contour · quadratic B(c)

MC2.05.21.24_ LaCl_3 · B-form: poly2 · profile contour ($L_p \times \sigma$, B(c) re-solved at each node)



Objective landscape per channel (mass | permeate | retentate | combined), other params re-solved at each node; red ▲ = minimum. A sharp minimum = identified; a flat valley along σ = the σ -rail.

→ shallow/flat objective — the model-limited case; the landscape itself shows poor identifiability.

What the four fits show

- NaCl (both regimes): constant B is sufficient — B is concentration-independent (Peclet $\rightarrow 0$, diffusion-limited for the diluting case). Mass and both concentrations are tracked at $WSSE_3 \approx 42 / 114$.
- $CaCl_2$: constant B fails (σ rails, $WSSE_3=220$). A quadratic $B(c)$ is required and cuts the error $\sim 33\%$ to 147 — consistent with convection+diffusion transport ($Pe \gg 1$) and the DATA2 per-response AIC selecting quadratic.
- $LaCl_3$: model-limited — every $B(c)$ form sits at $WSSE_3 \approx 1190-1244$; the residual points past $B(c)$ alone (e.g. $\sigma(c)$ / ion-pairing for the trivalent).
- Across the campaign: same NF270 membrane $\rightarrow L_p$ is consistent; the next step is the multi-experiment pooled fit (shared $\sigma + B(c)$) that resolves the per-experiment σ -identifiability.